Distributed and local computing on low-end devices from data privacy and decentralization perspectives

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***Abstract*—In the current geopolitical landscape, a limited number of dominant actors have the power to unilaterally block access to critical technologies or enforce systemic censorship. In response to this instability and concentration of control, security regulations within governmental and military organizations strictly prohibit the use of external services to ensure operational independence and prevent data leakage of classified information. Therefore, we developed the concept of a distributed network that focuses on decentralized computations and digital sovereignty, including both public and private hosting. Additionally, we collected a dataset of 1745 publicly known Android-compatible smartphone models to demonstrate the network applicability on low-end devices. Analysis results illustrate that approximately 66.78% (989 out of 1481) of the examined samples received their latest security patches three or more years ago. These statistics indicate that utilizing legacy devices as nodes in public networks is a significant security threat due to a lack of support and backports, compatibility issues, kernel vulnerabilities, and other limitations. Consequently, they should only be deployed with associated risks in locally isolated environments.**

Keywords—distributed systems, small models, mobile inference, low-end devices, decentralized network, data privacy, defense-in-depth, trusted and confidential computing

I. Introduction

The technofeudalism system of major corporations dictates conditions in which their cloud services eliminate individual ownership through vendor lock-in ecosystems and subscription-based models. Furthermore, these market conditions drive manufacturers and suppliers to operate as cartels, dictating prices and monopolizing across various enterprise segments. There are numerous preconditions that have formed this situation. Firstly, the race of artificial intelligence (AI) and associated information technologies demand a rapid expansion of computational resources in IT infrastructures. For example, Pilz et al. highlight a trend based on data from 2019 to 2025 years, showing that the computational performance of leading AI supercomputers doubles every 9 months, whereas their power requirements double every 1 year [1, Sec. 3.2-3.3]. Moreover, the total electricity capacity required even by a single leading AI data center may reach 9 GW (gigawatts) by 2030 in projections, given the continuing pace of growth [1, Sec. 4.1-4.2]. Collectively, these factors provoke market manipulations and stimulate growth of prices with shortages of critical resources (e.g., energy and raw materials, hardware components, qualified personnel, water for cooling, etc.). Secondly, the over-reliance on centralized data centers represents single points of failure that are vulnerable to physical damage or service disruptions caused by state policies. This is particularly critical given that statistics of 2025 show 75% of data centers are concentrated within the United States of America (USA), with the remainder distributed in other countries [1, Sec. 3.6]. Thirdly, many hardware technologies and their production remain strictly proprietary, resulting in a lack of competition between actors and reducing overall production rates. Specifically, the USA's national policy has already undergone three phases since 2018 for partial export bans on critical technologies targeting China and other market participants [2, Ch. 3]. Also, the industry faces various issues, including state-mandated backdoors, supply chain bottlenecks, planned obsolescence, and accumulation of electronic waste.

For these reasons, independent nations should pursue technological autonomy by developing local production in software and hardware without becoming hyper-dependent on untrusted partners, while still leveraging global experience and standards. Especially, military contexts of combat operations require portable and autonomous systems to mitigate the risk of lost or disrupted communications. Historically, this necessity drove the development of networks such as ARPANET and Usenet, born from the synergy of academic research and government-military collaboration [3].

Our research focuses on the discussed challenges, investigating how small-scale groups can compete in the AI sphere, combining cost-effective or low-power devices with data privacy without exposure to third parties. Firstly, the literature review covers three major topics: application of decentralized networks for general-purpose tasks (e.g., data storage, in-memory processing, and runtime execution); introduction of optimization techniques designed to democratize AI; analysis of general security aspects. Secondly, the methodology shows the concept of the proposed distributed network by covering its tokenomics, resistance, and fairness to prevent centralization of nodes in pools. Thirdly, the results describe our dataset of Android-compatible models of smartphones with their characteristics. Finally, we cover security-related issues for self-hosting users and local company infrastructures, including the absence of packet validation, old security updates, session and token-based authentication, mechanisms to mitigate data spamming, network constraints, and general limitations.

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II. Literature review

A. Distributed computing and blockchain-based systems

The development of the Internet and the advancement of networking technologies provided capabilities for computing systems to operate across geographically distributed locations, accelerating information exchange, the accumulation of collective knowledge, and hosting both public or private digital services [3, pp. 9–10]. For example, crypto networks as overlay networks and Web3 technologies allow to conduct direct financial transactions and peer-to-peer (P2P) connections between multiple nodes. Several of these decentralized networks leverage blockchain as an immutable data structure to ensure trust and data integrity [4, Ch. 3]. Besides, there are alternative architectures and structures (e.g., graphs, trees, unstructured ones, and so on [4, Ch. 6]) with other characteristics and without known limitations of blockchain:

- Versioning or migration during hard forks requires branching the chain. It remains a voluntary decision for network participants, as they may choose to stay on the legacy branch or migrate to the new one [5, pp. 1–2].
- Scaling of nodes requires off-chain data separation, as large blocks cause network congestion and degrades transaction performance [6, Ch. 2–3].
- Block propagation or relaying time requires prevention of redundant data transmissions and shorter distances to be constant, which is difficult to maintain across large distances between nodes when also dealing with cross-border deployments or regional restrictions [7].

Distributed systems are characterized by having consensus protocols and zero-knowledge proofs, which are both mechanisms for validating or verifying computational correctness and compliance with a predefined set of rules [8]. Their development and approbation are complex processeses involving rigorous vulnerability analysis and formal verifications [9]. For example, the framework proposed in [10] integrates proof assistants with generative language models to automate the mathematical verification of protocols. However, experts raise ecological concerns by shifting towards cheap renewable energy and energy-efficient optimizations, because certain consensus mechanisms such as Proof of Work (PoW) require intensive redundant computations that consume more power for protocol security than providing tangible utility for humanity [11], [12]. Therefore, [13] and [14] advocate for Proof of Useful Work (PoUW), which provides practical applications for AI model training, algorithmic optimization, data labeling, and other valuable tasks. Moreover, various factors within certain crypto networks (e.g., dynamic difficulty, halving events, and market hype) have shaped their economics to token holding and the centralization of power within large pools. For instance, FruitChains tries to mitigate it by guaranteeing fair reward distribution proportional to wasted computational power [15]. Also, Auditable Proof-of-Work (APoW) allows nodes to cryptographically prove their efforts, audit participants, and detect distributed attacks [16].

Centrally controlled and widely used cloud services (e.g., Amazon Web Services, Google Cloud, Microsoft Azure) can pose a significant risk to dependent infrastructure and users because their outages and failures can halt essential operations, as shown in the aggregated data of Talluri et al. from 2018 to 2020 [17]. The authors also state that these situations are frequent and natural occurrences ranging from minor to major ones, noting that a discrepancy between official reports and often-unreliable user feedback makes it more difficult to measure the full extent of such incidents [17, pp. 5–7]. Table 1 describes several examples of how distributed ledger mechanisms can decentralize traditional IT infrastructures to improve reliability and uptime while reducing the risk of single points of failure and dependence on centralized management. However, Balduf et al. daily crawled traffic and network participants in the IPFS (InterPlanetary File System), one of the largest decentralized P2P storage, during a single month period in 2023, showing that 20300 (79.6% of nodes) of approximately 30000 nodes across 152 countries were hosted on cloud-based VPS (Virtual Private Servers), with a notable concentration in the USA (33% of nodes) [18]. It reveals that despite having decentralized networks, they still remain deeply interconnected with the underlying centralized ecosystem, while there are clear and understandable reasons for this phenomenon:

- Difficulties of exposing a local node with public IP address due to the lack of IPv6 support and NAT (Network Address Translation) configuration from the ISP (Internet Service Provider) [18, p. 6].
- The limited number of IPFS gateways acts as bottleneck, with their resolution partially reliant on Cloudflare services [18, pp. 3, 11].
- Usage of cloud services for privacy and convenience.

TABLE I. Decentralized services over centralized ones

Authors	***Concepts***
Stoz and Debatty [19]	Decentralized architecture leveraging IPFS for software package management.
Ince et al. [20]	System for Linux ecosystem in which package providers act as nodes for reproducible building of software packages.
Costa et al. [21] and Scheid et al. [22]	Distributed repositories for open-source projects and software packages with P2P connections.
Hammad et al. [23] and Nizamuddin et al. [24]	Decentralized and immutable version control systems to secure the software development lifecycle.
Giamouridis et al. [25]	Blockchain-based DNS (BDNS) against bottlenecks of most DNS (Domain Name System) and CDN (Content Delivery Network) servers/services.

B. Optimizing computing and small models on edge devices

While historical trends indicate that computational demands have continuously increased since the inception of deep learning and throughout the current large-scale era of AI [26, Ch. 3], the continuous scaling paradigm is hitting a plateau. Physical constraints and diminishing returns make this approach increasingly unsustainable, requiring search for more extensive methods [27]. Experts propose various modern solutions and enhancements from both hardware and software sides. For instance, Li et al. demonstrated that offloading partial neural network inference or other computational tasks from data centers to edge devices can reduce environmental impacts (e.g., energy consumption, carbon emissions, and water usage) in their experiments [28]. Particularly, it is possible to construct a pipeline/workflow that evaluates the complexity of a given task and redirects it to a smaller model, achieving reasonable generation speed and sufficient output quality for specialized tasks or domains [29]: mathematical proofs, seq2seq (sequence to sequence), NER (named entity recognition for structured extraction of entities or actions), classifications, formatting and autocompletion of coding snippets, automating repetitive actions with agentic AI, searching and sorting unstructured data, and so on. These capabilities and efficiency of on-device processing are primarily caused by the SoC (System-on-Chip) architecture of mobile devices, which allows utilization of GPU (Graphics

Processing Unit), CPU (Central Processing Unit), and NPU (Neural Processing Unit) for tensor and matrix computations on a single chip [30]. Moreover, Amatuni et al. represent USA statistics in which the average lifetime/lifecycle of such devices can be extended by around 18% (from 2.7 to 3.2 years) through secondary markets, thereby reducing the production volume of new units [31]. However, the root cause of the problem arises from manufacturers and distributors who refuse to make their devices repairable, reliable, durable, and equipped with interchangeable components across product lines or generations for ease of maintenance and repairment. Researchers [32] and [33] discuss that open hardware and electronics can democratize high-performance computing, enhance reproducibility, and facilitate the sharing of experience for iterative refinement by applying RISC-V instruction set, reprogrammable FPGAs (Field-programmable Gate Array), Vulkan and OpenGL backend instead of proprietary CUDA, ASIC (Application-Specific Integrated Circuit) as special-purpose hardware, and customizable microcontrollers. From the software part, there should be a reduction of power consumption and environmental footprint that involves optimizing algorithms (following Big-O notation), adopting energy-efficient programming languages (compiled languages instead of interpreted ones), increasing hardware utilization (concurrency and parallelism), lower memory usage of a programming language (PL), and restarting programs on demand instead of background running [34]. Kempen et al. noted that specific implementations of PL with the same syntax can differ in their runtime, clearly showing that LuaJIT and PyPy outperform Lua and CPython by 500% and 125% in execution speed and energy efficiency respectively in their benchmarks, because JIT (Just-in-time) optimizer/compiler can simplify hotspots, loops, and frequent instructions [35, pp. 7–8].

DeepSeek, an independent Chinese AI laboratory, demonstrated that training a 671B (billion) parameter state-of-the-art (SOTA) large language model can be achieved with 2.788 million hours of NVIDIA H800 GPU (approximately 5.6 million USD at 2024 market rates) through the use of knowledge distillation techniques [36]. However, such estimates do not entirely represent the actual situation, as they do not account for associated costs such as:

- maintenance of IT infrastructure and hardware accounting (e.g., service lifetime, wear, and degradation) due to overheating or throttling under high-intensity loads or environmental hazards;
- employee salaries and fees for external services;
- expanding density and scale of digital information for multimodal capabilities (video, audio, images, etc.).

In this regard, federated learning allows to shift the responsibility for physical hardware maintenance from a centralized server to a distributed network of nodes that serve abstraction of resources for AI-related tasks such as training models, processing large datasets, running model inference, data annotations, and so on [37]. It is categorized by several sub-types on data organization and structures:

- Horizontal (same datasets but different samples), vertical (complementary features from overlapping samples), transfer learning (adapting a model trained on one dataset to perform a different but related task) [37, Ch. 2–3].
- Local (on a single device), decentralized (each node acts independently), personalized (selecting specific nodes based on their state or criteria), and centralized (as demonstrated in Fig. 1) [38, Sec. 2.1-2.3].

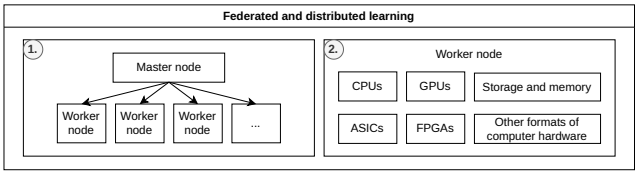

Fig. 1. Federated learning and distributed computations, adapted from [38, pp. 2, 6]: (1). Centralized form of federated and distributed learning that are constrained by the networking limitations of endpoints on data volume and communication speed. (2). A worker node operates on heterogeneous hardware such as microcontrollers, single-board computers, and other devices.

In general, federated learning is a much broader field, as demonstrated in Table 2, which briefly outlines various use cases and their specific characteristics.

TABLE II. CASES OF FEDERATED AND DISTRIBUTED LEARNING

Authors	*Concepts*
Li et al. [39]	This paper introduces the Adaptive and Resilient Model-Distributed Inference (AR-MDI) algorithm to address the limitations of traditional data-distributed inference (DDI). While DDI requires each worker to store a full pre-trained model and involves high communication costs for large data, AR-MDI partitions the Deep Neural Network (DNN) across multiple workers.
Beltrán et al. [40]	This comprehensive survey examines the transition from Centralized Federated Learning (CFL) to Decentralized Federated Learning (DFL) by promoting decentralized model aggregation and categorizing DFL into various architectures (cross-silo, cross-device) and topologies (fully or partially connected).
Cajaraville-Aboy et al. [41]	They propose a multi-domain orchestration architecture for Fluid Computing, treating heterogeneous resources (edge, fog, cloud) as a unified orchestration plane. Also, they developed an SDN (Software-defined networking) anomaly detection mechanism which is designed to enhance security against Byzantine threats (malicious or failing nodes) in a decentralized environment.
Macario et al. [42]	This research presents MDI-LLM, a framework for deploying Large Language Models (LLMs) by pooling resources from multiple low-power edge devices, in which the number of participants boosts overall token-generation throughput and reduces the memory footprint per device.
Li and Teng [43]	The authors formulated a priority-aware model allocation optimization problem for MDI (PA-MDI) to determine how model layers should be distributed across devices based on priority and levels. For example, their benchmarks on the NVIDIA Jetson Xavier demonstrated reduced inference times for high-priority tasks with the ResNet-50 and GPT-2 models.
Hard et al. [44]	This paper demonstrates training process of recurrent neural networks for next-word prediction in Google's Gboard by applying federated learning. It highlights that training on local device caches provides access to higher-quality datasets that more accurately reflect real-world typing distributions than truncated server-hosted logs.
Telegram [45]	COCOON (Confidential Compute Open Network) is a decentralized multimodal AI inference network which is integrated into Telegram social platform and their TON blockchain. It rewards participants for their GPU contributions in real-time, while ensuring scalability and censorship resistance.
Matsutani et al. [46]	The authors propose a distributed prompt caching mechanism that allows multiple low-end devices to cooperatively share intermediate processing states (key value caches) through a central remote server. Each client has a Bloom-filter-based catalog to determine if the required states exist on the server before attempting a download for mitigating overhead of wireless communications. Testing with the Gemma-3 270M model on Raspberry Pi Zero 2W hardware showed that their approach reduces Time to First Token (TTFT) by 93.12% and Time to Last Token (TTLT) by 50.07% on average.

C. Security and privacy of execution environments

Federated learning can also handle privacy aspects by having each participant share only a snapshot of their locally trained model (its insights in model weights and intermediate activation vectors) to aggregate and combine them via secure multi-party computation, rather than transferring raw datasets directly [47]. In the context of medical records, it is mandatory to implement data anonymization, model encryption, and differential privacy through noise injection [48]. Also, it is even possible to facilitate ephemeral data access, in which data is temporarily cached in memory for processing and then immediately cleared [47, p. 11]. Moreover, certain networks like Monero are strictly focused on participant privacy, private payments, private computations, and function privacy [49, pp. 7–8].

Security in distributed systems is critical for stabilizing nodes and preventing localized issues from propagating across

the network. Daulay et al. investigated the security risks of inference attacks aimed at extracting sensitive data from model weights and disrupting network integrity (e.g., data spamming), noting that the inherent complexity of distributed governance makes ensuring honest participation and data protection particularly challenging [50]. Therefore, there is the concept of Trusted Execution Environments (TEEs) that introduces:

- Hardware and software isolated execution restricts operations to an isolated environment, functioning as both a trusted and controlled environment [51].
- Mitigation of memory corruption, various vulnerabilities, and low-level kernel exploits by utilizing memory safety and specific modifications [52].
- Confidential computing for sensitive operations ensures that network participants remain auditable, protected, and verifiable [53].
- Application of TPM (Trusted Platform Module) chips for hardware-based trust, MAC (Medium Access Control) addresses for networking hardware identification, and specialized firmware to ensure the integrity of nodes [54].

By default, mobile operating systems such as Android already offer robust and secure-by-default capabilities through restrictive application sandboxing and a strict permission model [55]. Nevertheless, malware detection and dependency analysis still remain essential for evaluating untrusted applications and third-party libraries [56]. Chen et al. evaluated WebAssembly (WASM) as a viable execution runtime that provides a secure and platform-agnostic sandbox [57]. Also, experts suggest using WASM for smart contracts, where programs are compiled or encoded into WASM bytecode [58, Ch. 3]. Although bad actors may exploit distributed networks for malicious purposes (e.g., Command and Control (C2) infrastructure, illegal operations, or bypassing protective measures to abuse resources), researchers propose methods for preserving the computational correctness and moderation of smart contracts by using fuzzers or scanners to identify vulnerable or malicious patterns [59].

III. Methodology

A. *Concept of a distributed network*

We specified mechanisms of consensus and rules by which network participants, as nodes, reach agreement or are configured for correct operations. Our network is based on PoUW for solving useful computing and other defined tasks. There is a primary abstraction termed a “unit of work” that is achieved by completion of relatively “valuable or useful work”. The network has two roles of participants who determine the work’s usefulness and its intrinsic value:

- Providers can create tasks by publishing them to the network and determining the validation criteria for received results.
- Contractors can select available tasks from the network and perform/execute them.

During the prototyping and early stages of the system, we consider a limited set of predefined/fixed types of tasks, including data annotation, dataset storage, model inference, and model training. Future iterations of the network will support custom general-purpose and computational tasks that are defined and implemented on the software side via smart contracts on WASM:

- ASIC-resistance due to the diversity of tasks that cannot be solved using a single method/algorithm.
- Preserving the value of the completed work, compliance with specified criteria, and other features.
- Providing additional packages and functionalities (e.g., visual modifications) for certain specifics of a task.

Fig. 2 demonstrates that our consensus is an intermediate solution that varies dynamically from task to task. Also, it is possible to combine PoW and PoUW in particular proportions to maintain baseline security guarantees and ensure a seamless transition from legacy networks (e.g., a distribution of 20% for PoW and 80% for PoUW).

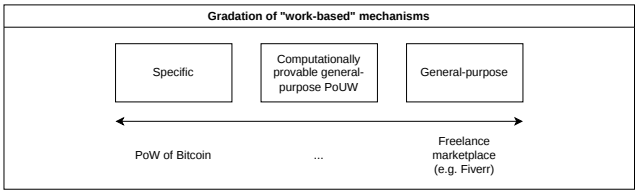

Fig. 2. Gradation of work-based consensus mechanisms: Specific one (on the left) is based on a single fixed algorithm or task type. General-purpose one (on the right) is decoupled from computing power, so it is too abstract and difficult to verify. Our PoUW (on the center) is based on computationally provable tasks, but also allows custom general-purpose tasks on demand.

For example, if organizations need to train an AI models or store public datasets without maintaining their own IT infrastructures, they can acquire tokens to request on-demand computing power by publishing a smart contract that broadcasts/emits their requirements to the network. Then, someone can accept or join the task in a similar way to quests in multiplayer games, as illustrated in Fig. 3. Contractors can choose tasks manually or enable an automatic mode to match them based on a specific algorithm or preferences. Alongside tasks from individuals, partner-sponsored storylines and collective missions would provide a continuous stream of work for participants during periods of reduced network activity.

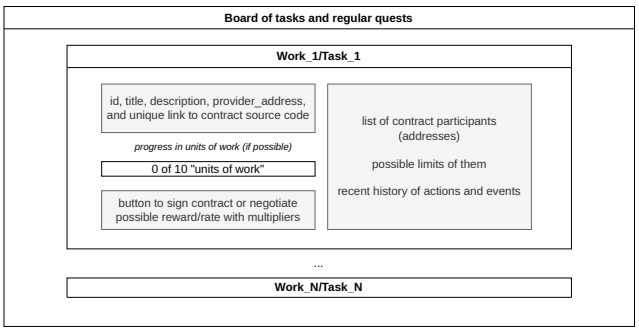

Fig. 3. Market-driven mechanisms for providers and contractors. In this context, tasks are valued relative to a set rate per unit of work in tokens.

Furthermore, we propose a concept of fairness, which is achieved through a limited total supply of tokens and an inflation mechanism designed to prevent holding and market monopolization:

- If tokens are not spent or transferred within a specific timeframe, they gradually burn from the wallet, returning to the limited total supply, as illustrated in Fig. 4.
- A single wallet has a maximum limit of tokens.

- Network participants have the option to voluntarily burn some amount of tokens to refill total supply.

Initial tokens in the network is generated via a Genesis Task, which handles minting and system bootstrapping, as otherwise token acquisition would be impossible.

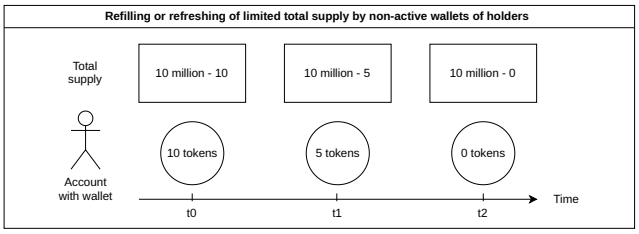

Fig. 4. Refilling of limited total supply by non-active wallets of holders or by burning tokens from their wallets.

From a technical standpoint, we want to build upon the foundations of IPFS and Ethereum, while introducing proprietary extensions and enhancements. As shown in Fig. 5, a node is any single component within the network that executes our certified cross-platform software (daemon, browser-based, or native) and establishes connections by adhering to the consensus protocol with underlying networking stack. Especially, IPFS has libp2p as its core library, which can be reused for our system, as it provides:

- DHT (Distributed Hash Table) for data storage management across various network topologies (e.g., mesh, ring, and hybrid).
- Bitswap algorithm for P2P routing and resource/content discovery among the nearest neighboring nodes.
- Other modifiable features and implemented protocols that are essential for building decentralized networks.

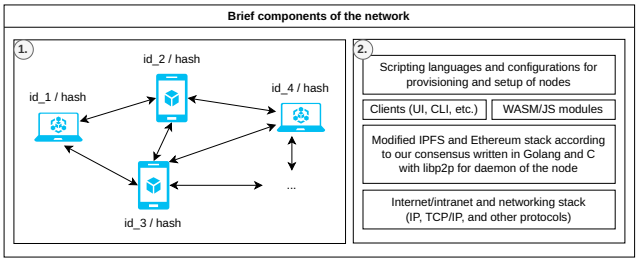

Fig. 5. Components of the overlay network: (1). Visualization for portion of heterogeneous nodes with unique identifiers or hashes in the P2P network. (2). Separation of concerns with technical stack.

On top of these foundations, there are layered ecosystem to separate users based on their proficiency and interests, as illustrated in Fig. 6. The first layer is a rich-client wrapper with auto-updates featuring gamification elements and interactive modes, in which published tasks and actions are seamlessly integrated into mini-games and quests. It facilitates active token farming and rewards from in-game assets (e.g., skins, models, and boosts) with social interactions to foster community engagement through private and public chats. The second layer focuses more on technically proficient participants by offering:

- Passive utilization of computing resources operating as a legitimate miner with a slider for manual load adjustment or automatic dynamic tuning based on available resources. For instance, the system calculates the necessary overhead while maintaining a base cap of approximately 20% of total device load to prevent hardware wear or excessive battery drain.
- Developing custom smart contracts and hosting decentralized services, such as VCS (Version Control System) hosting of repositories and package registries with distributed CI/CD pipeline execution, transparency, and immutable history.

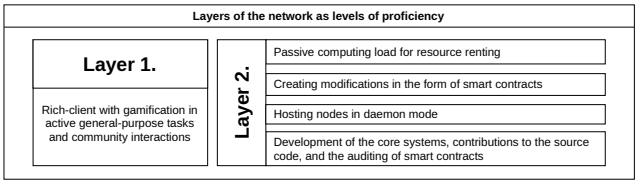

Fig. 6. Layers of the network as levels of proficiency.

As shown in Fig. 7, the system allows connections to various subnets, offering a choice between the main public realm (which requires an internet connection) and alternative subnets. This architecture loosely resembles the mechanics of Usenet groups, which organized multiple nodes based on shared interests or specific characteristics. Furthermore, to ensure fault tolerance against system crashes and data transmission issues, data is backed up to a persistent SQLite database and automatically synchronized upon reconnection.

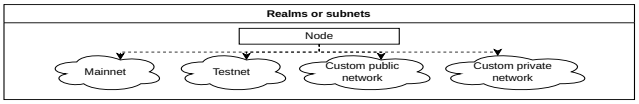

Fig. 7. Nodes can configure connections to specific realms or subnets by specifying the network's RPC (Remote Procedure Call) URL and associated metadata.

B. *Practical analysis of Android-compatible devices*

We gathered a dataset from public sources consisting of Android-compatible smartphones ranging from Android version 1 to 16 and beyond, containing 1745 unique models after pre-processing. The primary objective was to study security issues, unique features, and general complexities of utilizing such devices as low-end nodes. Firstly, detailed description of the dataset is provided in Table 3. Secondly, data collection was carried out manually and through Python scripts. Thirdly, visualizations were rendered in Matplotlib and processed using Pandas, while the raw data was saved in a structured JSON format. Finally, the remaining details are described in more depth in the repository containing our source materials [60].

TABLE III. DESCRIPTION OF DATASET WITH ANDROID SMARTPHONES

Field	*Data type and description*
model	str; the formal or consumer name of the device model
developer	str; the manufacturer or vendor that developed, sells, or brands the specific device instance
release_date	datetime; the date of public release for consumer purchase
android_version	str; the stock version at release or the default minimal version supported by the smartphone in normalized form (“major.minor.patch”)
status	str; one of the possible statuses for the smartphone’s Android version (latest, unknown, supported, or unsupported)
latest_security_patch_date	datetime; the date of the latest / last security patch released for the smartphone's Android version

IV. RESULTS AND DISCUSSIONS

A. *Analysis results and statistical charts from the dataset*

Fig. 8 represents the major Android versions across the collected dataset of smartphones, where popularity peaks are visible at versions 4, 10, and 11. Unclear decline is also noticeable starting from version 11, but it is most likely related to the limited sample size in our dataset, possible drop in

interest in Android devices, a reduction in smartphone model segments, monopolization around a limited number of manufacturers, or limited support for the latest Android versions from custom distributions (e.g., MIUI, OneUI, etc.). The graph illustrates that there are 980 devices with Android versions ranging from 1 to 9 out of the 1742 shown in total, indicating a massive volume of old or legacy devices that require attention in order to find a practical use for them.

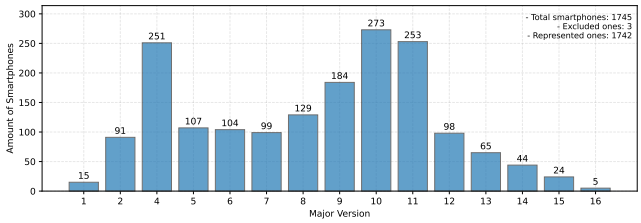

Fig. 8. Major Android versions across smartphone models, excluding certain smartphones due to inconsistencies in their versioning semantics and unconfirmed / unknown Android versions on them.

Fig. 9 groups devices by their versions relative to support for Termux which is a terminal emulator for Android that allows running ported binaries, scripts, or acting similar to Linux distribution environment, rather than creating a separate application with calls to the Android API. Notably, the majority of devices of 67.4% fall into the compatible category that we are targeting, while the remaining 32.6% represent a problematic segment that is difficult to work with and maintain.

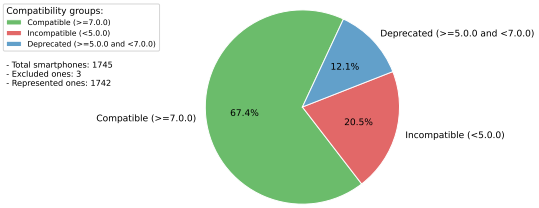

Fig. 9. Compatibility with Termux depending on Android versions (% of smartphones). Deprecated status formally means that it works on older versions without the Termux package ecosystem, which requires building packages from source. Other incompatibles might be mitigated by manually porting Termux with wrappers over the Android API for specific old versions, but without proper support or patches for kernel vulnerabilities.

Fig. 10 and 11 address security and the level of support these devices receive in terms of Android version and kernel updates. The Android version status for 1669 devices is "unsupported", meaning the version is not the latest and has reached end-of-life which is largely applies to all versions except 15 and 16. Of course, this does not rule out the existence of backports from Google when necessary, nor does it exclude support models provided by manufacturers for custom firmware. Fig. 11 expands on this understanding by detailing critical facts regarding smartphone releases, with a peak in 2019 and 2020. Furthermore, the security aspect is examined from the perspective of the latest security patch, which reveals that approximately 66.78%, amounting to 989 out of 1481 devices, received their latest security patches for their Android version three or more years ago.

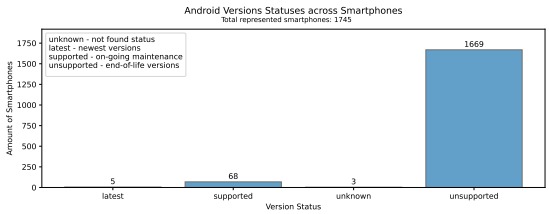

Fig. 10. Device statuses, where some are displayed as “unknown” because the Android versions on them were unidentified.

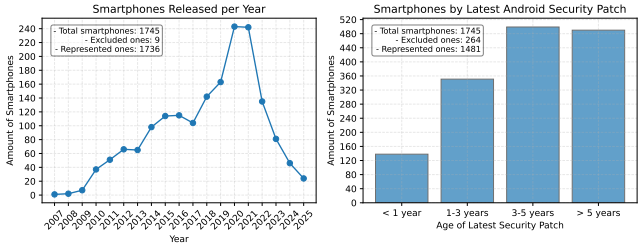

Fig. 11. Amount of smartphone models released per year and their latest Android security patches.

B. *General issues and state of security*

The utilization of legacy Android smartphones for a distributed network is a complex set of physical and systemic challenges. Because of outdated firmware and unsupported Android versions, these devices are vulnerable to unpatched exploits at both the operating system and kernel levels, requiring significant technical expertise to mitigate through methods such as rooting or installing custom distributions. Consequently, connecting them to the public web poses a substantial security risk. It is only possible to use them in isolated local environments and private networks. Also, these devices struggle to maintain a charge under load for extended periods due to severe battery wear and overall hardware degradation. In some jurisdictions, utilizing such devices is prohibited under European Union energy standards because they are potential fire hazards and are significantly less efficient, resulting in higher power consumption with lower performance output.

The network still faces other unresolved and ambiguous issues. Firstly, the proof-of-burn mechanic may discourage certain participants due to the permanent loss of accumulated wealth. Furthermore, scenarios may arise where the total supply is depleted while existing tokens remain unburned, resulting in a liquidity crisis for new participants due to the limited overall token availability. Our initial intent was to incentivize the constant circulation of capital rather than passive holding, thereby mitigating the influence of large-scale stakeholders over network control. Secondly, we had ideas to explore an investment mechanism involving registered public entities, such as universities or charities. For example, users could delegate tokens to these organizations to defer the burn rate for a specified period. However, it is not sustainable due to potential exploitation schemes, such as the use of cryptocurrency tumblers and mixers to cycle tokens between accounts, or the deployment of empty contracts by bot farms to manipulate network activity. Thirdly, social and trust factors remain unavoidable, as actors may intentionally or unintentionally engage in activities such as providing low-quality results (empty data labels, fake model inference, incompleted submissions, and different disruptions by repeating actions as a form of distributed denial-of-service attack), organizing collusions, and other malicious schemes. So, the network may need external validators for quality assessment and moderation, as well as collaboration with local government authorities to implement user verification through the proof-of-identity and the enforcement of more restrictive operational limits.

V. Conclusion

Repurposing legacy smartphones to build decentralized networks during hardware shortages remains a critical goal, though many older devices risk becoming electronic waste due to poor performance and a lack of crucial security updates. The possible success of this initiative depends heavily on its technical execution and whether it can offer a genuinely

useful, viable, and feasible alternative to current centralized systems. At this stage, the project serves as an experimental foundational draft, meaning its precise operational mechanisms have not yet been fully defined. These complex system dynamics and actor interactions require ongoing, iterative research and refinement well beyond the scope of a single introductory article. Furthermore, significant technical hurdles remain unresolved, particularly concerning strict Android security policies that often require device verification and block the execution of uncertified applications.

Future work will prioritize essential auxiliary infrastructure, specifically the construction of custom-engineered device farms using modular, 3D-printed server racks equipped with active cooling and continuous power to prevent overheating and rapid battery depletion. Additionally, we plan to systematically gather detailed data across various low-end device categories to establish precise performance benchmarks, hardware characteristics, and application compatibility profiles. There is also a significant opportunity to enhance the network's AI capabilities by implementing self-configuring, distributed inference loops and meta-prompting via tools like llama-cpp and llama-bench, optimized specifically for low-resource hardware. Beyond our primary system architecture, these repurposed smartphones can support broader practical applications by leveraging Termux, proot-distro for Debian packages, and Android Debug Bridge (ADB) connections. However, successfully porting such applications to this ecosystem remains a notable challenge, largely due to the inherent technical difficulties of running systemd natively on Android, even when utilizing workarounds like termux-services. For example, we consider different applications:

- Custom inference servers by using the llama-cpp engine to host localized language models, utilizing RPC to distribute computational workloads across the device network.
- Decentralized storage and hosting by integrating IPFS network for persistent data storage of open-source repositories, alongside distributed web hosting that serves resources via individual devices using a load balancer.
- General distributed computing for execution of general-purpose tasks, including cryptocurrency mining operations using specialized software like cpuminer connected through a stratum server.

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